✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 5

Doppler Radar Systems

Radio Navigation & Aids — Ground Training Series

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1. Introduction

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What this section covers: The purpose of a Doppler navigation system, how it fits into modern avionics, and where the Doppler principle appears across different aviation systems.

A **Doppler navigation system** uses the Doppler principle to measure an aircraft's **ground speed** and **drift**. Combined with a heading input, these values allow a navigation computer to determine the aircraft's position — continuously updated without external signals.

🔵 **Where Doppler Is Used in Aviation**

- **Doppler Navigation Systems** — self-contained ground speed and drift measurement
- **Doppler VOR (DVOR)** — uses Doppler shift to produce the bearing signal
- **VDF (VHF Direction Finder)** — Doppler technique for bearing determination
- **Police radar** — classic real-world example of Doppler speed measurement
- **Weather radar** — Doppler shift detects precipitation movement and wind shear

Modern systems combine Doppler measurements with **IRS (Inertial Reference System)**, **VOR/DME**, or **GPS** in various configurations. This integration overcomes the key limitations of early Doppler systems: inaccurate heading references and signal degradation over large expanses of water.

2. The Doppler Principle

What this section covers: The physics of the Doppler effect explained by analogy, the exact relationship between relative motion and frequency shift, and the definition of Doppler frequency.

The Doppler effect was predicted by the Austrian physicist **Christian Doppler** in the 19th century for light waves — it applies equally to sound waves and radio waves.

> *"A received frequency will only be the same as the transmitted frequency when there is no relative movement between the transmitter and receiver."*

The Beach Analogy

Imagine standing still in the sea: waves strike you at a steady 4 per minute.

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- Walk **into** the waves → you reduce the distance between waves → they strike you **more frequently** (perceived frequency **increases**)
- Walk **away** from the waves → you increase the spacing → they strike you **less frequently** (perceived frequency **decreases**)
- The actual wave rate (4/min) has **not changed** — only your perceived rate has changed

Moving TOWARD Source



Received frequency INCREASES

Wavefronts are compressed
Shorter apparent wavelength
Higher perceived frequency

$f_{rx} > f_{tx}$

Moving AWAY From Source



Received frequency DECREASES

Wavefronts are stretched
Longer apparent wavelength
Lower perceived frequency

$f_{rx} < f_{tx}$

DOPPLER SHIFT (DOPPLER FREQUENCY)

$$f_{Doppler} = f_{received} - f_{transmitted}$$

$f_{Doppler}$ is proportional to relative velocity between transmitter and receiver

Approaching → $f_{Doppler}$ positive (received > transmitted)

Receding → $f_{Doppler}$ negative (received < transmitted)

Key Principle — Proportionality

The Doppler shift is **directly proportional** to the relative velocity. Double the speed → double the Doppler shift. This proportionality is what makes Doppler useful for measuring speed — a larger frequency shift means a higher relative velocity.

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3. Airborne Doppler — System Description

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What this section covers: How airborne Doppler radar is installed on an aircraft, the antenna type, the beam configuration, and the origin of the "Janus" name.

A typical airborne Doppler installation uses a **slotted waveguide antenna** in which:

- The transmitter and receiver elements are **screened from each other** but **share the same aerial**
- An array of beams is transmitted **downward** toward the Earth's surface

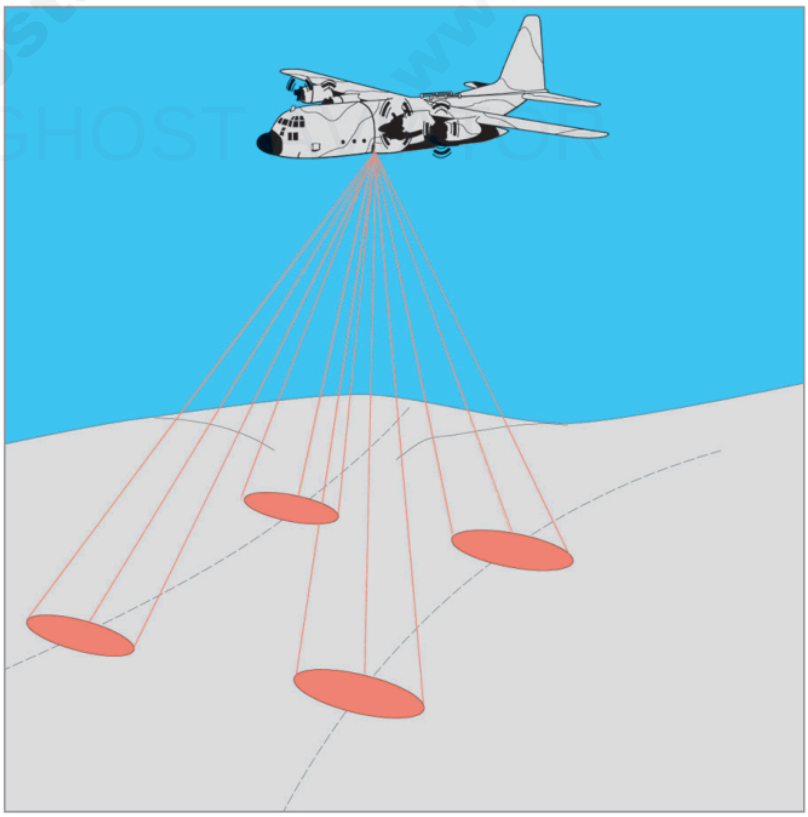


Figure 5.1 Airborne Doppler.

Figure 5.1 — Airborne Doppler installation: 4-beam Janus array with two beams pointing forward-left and forward-right, two pointing aft-left and aft-right. All beams are directed downward toward the ground. The aircraft's motion causes different Doppler shifts in each beam

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🔵 **The 4-Beam Janus Array**

- **4 beams total:** 2 pointing forward (port & starboard), 2 pointing aft (port & starboard)
- All beams directed **downward** at an angle to the Earth's surface
- Named after **Janus** — the Roman God of Doorways, who could face both directions simultaneously
- The forward/aft symmetry is what allows cancellation of aircraft pitch and roll errors

4. Janus Array Configurations

What this section covers: The various beam configurations used in Janus arrays — 3-beam and 4-beam layouts.

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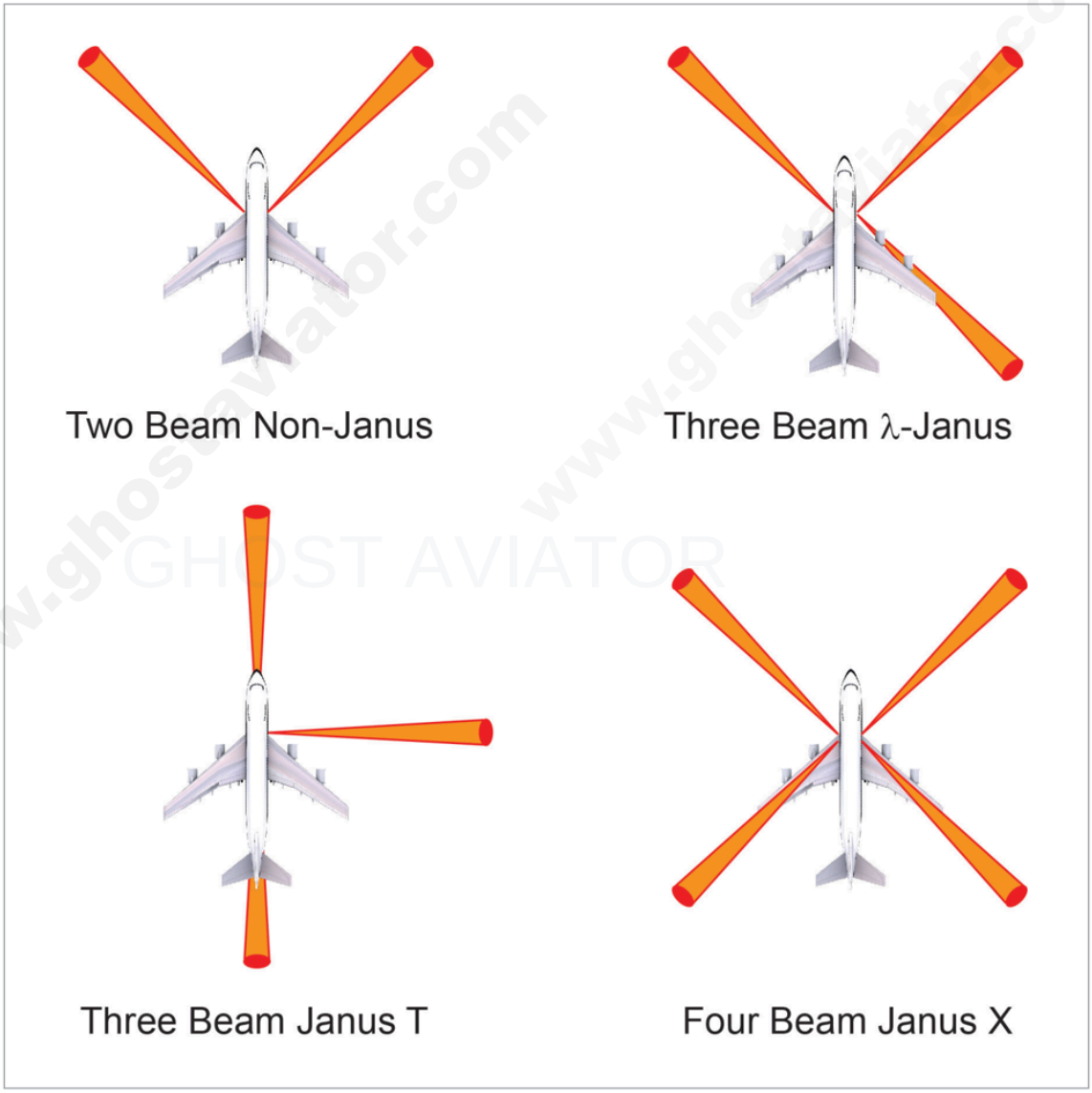


Figure 5.2 Janus Arrays

Figure 5.2 — Janus array configurations: a 3-beam system uses one forward beam and two aft beams at angles (or vice versa); a 4-beam system uses two forward and two aft beams symmetrically arranged. Various configurations trade off accuracy, complexity, and reliability

🔵 3-Beam vs 4-Beam Janus Array

- **3-beam:** typically one fore beam + two aft beams (or vice versa); simpler, lighter; less redundancy
- **4-beam:** two fore + two aft beams; better averaging; improved accuracy; standard configuration for most systems
- All Janus arrays have both forward-facing and aft-facing beams — the defining characteristic

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5. Doppler Operation — How Ground Speed & Drift Are Measured

> **What this section covers:** How the frequency shifts in the four beams are processed to give ground speed and drift, and how the system worked mechanically vs electronically.

The Doppler system **continuously** measures the frequency shift in the reflected signal caused by the aircraft's motion over the ground, then converts these values to:

- **Speed along track** → ground speed
- **Speed across track** → used to determine drift

Measuring Ground Speed

✅ **Forward vs Aft Beam Frequency Shifts**

- **Forward beams:** aircraft moving toward the reflected ground → received frequency is **HIGHER** than transmitted (positive Doppler shift)
- **Aft beams:** aircraft moving away from the reflected ground → received frequency is **LOWER** than transmitted (negative Doppler shift)
- With zero drift: forward and aft shifts are **equal in magnitude, opposite in sign**
- The magnitude of the shift is **proportional to ground speed**

Measuring Drift

✅ **Port vs Starboard Beam Frequency Differences**

- If aircraft drifts **left (port)**: port beams receive a different shift than starboard beams
- If aircraft drifts **right (starboard)**: starboard beams receive a different shift
- The **difference in frequency shifts between port and starboard beams** gives drift magnitude and direction

4 Janus Beams (Ground reflections)

Forward Beams	Aft Beams	Port/Stbd Difference
f received > f transmitted	f received < f transmitted	

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(+ve Doppler) (−ve Doppler)

Ground Speed
∝ magnitude of shift

Drift
∝ port vs starboard difference

↓ Navigation Computer: Position = f(GS, Drift, Heading)

Modern vs Earlier Systems

Feature	Modern (Fixed Aerial)	Earlier (Mechanical)
Aerial type	Fixed (no moving parts)	Pitch-stabilized, rotating aerial
Processing	Electronic processing of frequency differences	Motor-driven alignment until port = starboard shifts
Drift measurement	Computed electronically from beam differences	Pick-off measures heading vs aerial track alignment
Reliability	Higher (no mechanical wear)	Lower (mechanical complexity)

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6. Doppler Navigation Systems

> **What this section covers:** How Doppler measurements are converted to aircraft position in early and modern navigation computers.

The Doppler system continuously updates drift and ground speed values. A navigation computer uses three inputs to compute position:

DOPPLER NAVIGATION POSITION FIX

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Position = f(Ground Speed, Drift, Heading)
Output: Latitude / Longitude
or: Along-track / Across-track distance from departure point
```

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Figure 5.3 Racal RNS 252 Navigation Computer Unit

Figure 5.3 — Racal RNS 252 Navigation Computer Unit (CDU): the control and display unit from the B-52 Doppler navigation system, showing the typical interface for entering departure position and displaying computed position as along/across track distances or lat/long coordinates

🔵 **Navigation Computer Inputs & Outputs**

- **Inputs:** Ground speed (from Doppler fore/aft beams) + Drift (from port/stbd beams) + Heading (from compass/IRS)
- **Starting position:** Aircraft departure coordinates manually loaded
- **Outputs:** Current position as latitude/longitude *or* as along/across track distances from departure
- **Modern integration:** Doppler + IRS + VOR/DME + GPS for improved accuracy and redundancy

7. Doppler Limitations & Modern Integration

What this section covers: The two main limitations of early Doppler systems and how modern multi-sensor integration overcomes them.

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⚠ **Limitations of Early Doppler Navigation Systems**

1. **Inaccurate heading reference:** Doppler gives ground speed and drift but needs an external heading source. Early magnetic compasses introduced errors. Modern systems use IRS.
2. **Degradation/loss over water:** The smooth sea surface reflects Doppler signals differently from land — specular (mirror-like) reflection over calm water returns very little energy, causing signal degradation or complete loss. Wind-roughened sea improves this.

✅ **Modern Solutions**

- **Heading:** IRS (Inertial Reference System) provides accurate heading with no magnetic error
- **Over water:** Multi-sensor integration — GPS/VOR/DME provides position updates when Doppler signal is lost or unreliable
- **Integration modes:** Doppler + IRS; Doppler + GPS; Doppler + VOR/DME; fully integrated FMS solutions

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⚡ QUICK REVISION SUMMARY — CHAPTER 5

- **Doppler shift:** moving *toward* → frequency ↑; moving *away* → frequency ↓; shift ∝ relative velocity
- **Doppler frequency** = $f_{received} - f_{transmitted}$
- **Airborne Doppler:** slotted waveguide antenna; transmitter & receiver screened but share aerial
- **4-beam Janus array:** 2 fwd + 2 aft beams; named after Roman god Janus (faces both ways)
- **Ground speed:** magnitude of fore/aft Doppler shift
- **Drift:** difference between port and starboard beam frequency shifts
- **Zero drift:** fore/aft shifts are equal and opposite
- **Inputs to nav computer:** ground speed + drift + heading → position
- **Limitations:** heading accuracy; signal loss over calm water (specular reflection)
- **Modern systems:** integrated with IRS, VOR/DME, GPS
- **Also uses Doppler:** police radar, Doppler VOR, VDF, weather radar

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📝 Practice Questions & Detailed Answers

Q1. Doppler operates on the principle that _____ between a transmitter and receiver will cause the received frequency to _____ if the transmitter and receiver are moving _____.

a. apparent motion, decrease, together

b. relative motion, decrease, apart

c. the distance, increase, at the same speed

d. relative motion, increase, apart

✓ Correct Answer: **(b)** — relative motion, decrease, apart

Explanation: The Doppler effect occurs due to **relative motion** between transmitter and receiver. When they are moving **apart**, the received frequency **decreases** (wavefronts are stretched, longer apparent wavelength, lower perceived frequency). See Section 2.

Why the other options are wrong:

- **(a)** — "apparent motion" is vague/wrong; "together" is wrong direction for a decrease. Moving together (toward each other) causes an *increase*, not a decrease.
- **(c)** — "the distance" is imprecise; "at the same speed" → if transmitter and receiver move at the same speed in the same direction, *no* Doppler shift occurs (no relative motion).
- **(d)** — "relative motion" is correct, "apart" is correct, but "increase" is wrong. Moving apart causes a frequency *decrease*, not increase. This is the mirror-image trap.

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> ***Instructor's Note:*** *The trap in this question is option (d) — it gets "relative motion" and "apart" right but flips the frequency effect. Moving apart = frequency decrease (like ambulance siren fading). Moving together = frequency increase (like approaching siren). Internalize the analogy before the exam.*

Q2. Due to 'Doppler' effect an apparent decrease in the transmitted frequency, which is proportional to the transmitter's velocity, will occur when:

a. the transmitter and receiver move towards each other

b. the transmitter moves away from the receiver

c. the transmitter moves towards the receiver

d. both transmitter and receiver move away from each other

✓ Correct Answer: **(b)** — the transmitter moves away from the receiver

Explanation: A frequency **decrease** occurs when there is **increasing separation** — i.e. the transmitter moves *away* from the receiver (or vice versa). This stretches the wavefronts, reducing the perceived frequency. Note option (d) would also produce a decrease but is more complex — (b) is the cleaner, direct answer covering the fundamental case. See Section 2.

Why the other options are wrong:

- **(a)** — moving toward each other produces a frequency *increase*, not decrease.
- **(c)** — transmitter moving toward receiver produces a frequency *increase*. This is the direct opposite of what is asked.
- **(d)** — both moving away from each other would also produce a decrease, but the key difference is the *relative velocity*. The proportionality to "transmitter's velocity" in the question refers to the single-body case of (b), making (b) the best answer.

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> ***Instructor's Note:*** *Q1 and Q2 both test the same fundamental rule from different angles. Master this: toward = increase; away = decrease. The Doppler frequency is proportional to the component of relative velocity along the line between transmitter and receiver.*

Q3. The change in frequency measured in an aircraft from a radio transmission reflected from the ground is used to determine:

a. the drift and ground speed of the aircraft

b. the aircraft's track and speed

c. the across track wind component and heading

d. track error and ground speed

✓ Correct Answer: **(a)** — the drift and ground speed of the aircraft

Explanation: The Doppler system converts measured frequency shifts to: (1) **ground speed** — from the magnitude of the fore/aft beam frequency shifts; (2) **drift** — from the difference between port and starboard beam frequency shifts. Heading must be supplied separately. See Section 5.

Why the other options are wrong:

- **(b)** — Doppler does *not* directly measure track. Track = heading ± drift; it requires a heading input. Ground speed is not the same as "speed" (which is ambiguous). (a) is more precise.
- **(c)** — Doppler does not measure wind component or heading directly. Wind can be derived from the ground vector but is not the primary output. Heading is an external input.
- **(d)** — "Track error" is not a standard Doppler output. Doppler gives drift (the angle between heading and track), not track error per se. Ground speed is correct but the full answer (a) is more accurate.

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> ***Instructor's Note:*** *The precise outputs of an airborne Doppler system are: ground speed and drift. Track requires adding drift to heading (external input). Position requires ground speed + drift + heading + time. The textbook is explicit: "The equipment converts the measured values into the aircraft's speed along track (ground speed) and speed across track (used to determine drift)."*

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📊 Master Reference Tables

Doppler Shift Summary

Relative Motion	Frequency Effect	Wavefront
Moving toward each other	**f received > f transmitted** (increase)	Compressed (shorter λ)
Moving away from each other	**f received < f transmitted** (decrease)	Stretched (longer λ)
No relative motion	f received = f transmitted (no shift)	Unchanged

Janus Array — Beam Frequency Shifts

Beam	Condition	Frequency Shift	Derives
Forward beams	Aircraft moving forward	**+ve (increase)**	Ground speed
Aft beams	Aircraft moving forward	**−ve (decrease)**	Ground speed
Port vs Starboard	Aircraft drifting	**Port ≠ Starboard**	Drift
Port vs Starboard	Zero drift	Port = Starboard	Zero drift confirmed

Mnemonics & Memory Aids

- **Toward → Up:** Moving toward source → frequency UP (like approaching siren)
- **Away → Down:** Moving away → frequency DOWN (like receding siren)
- **Janus = Roman god of doorways** (faces both forward and back simultaneously)
- **Ground speed:** fore/aft shift magnitude; **drift:** port/stbd difference
- **Nav computer needs:** G/S + Drift + Heading → Position
- **Limitations:** Heading (→ use IRS); Over water (→ specular reflection loss)
- **Doppler used in:** Police radar, Doppler VOR, VDF, weather radar, AWR

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